

EDUCATION

- 2018.3 – 2023.2 **Ph.D. in Medicine (Applied Bio-Engineering)**, Seoul National University
Thesis: *Clinical Information Extraction from Unstructured Free-texts for Pharmacovigilance* · Advisor: Howard Lee
- 2013.3 – 2018.2 **B.S. in Biology and Anthropology**, Seoul National University — Summa Cum Laude

EXPERIENCE

- 2026.1 – Present **Sr. AI Engineer**, Selta Square
- 2024.8 – 2025.12 **Research Assistant Professor**, Biomedical Research Institute, Seoul National University Hospital
- 2023.3 – 2024.7 **Post-doctoral Researcher**, Seoul National University Hospital

RESEARCH INTERESTS

Clinical / Document Information Extraction · Biomedical Natural Language Processing · ML-based Drug Development · Clinical Informatics · Clinical Pharmacology & Regulatory Science

PUBLICATIONS

Journal Articles

- Improving clinical data standardization in surgical notes through biomedical entity linking with contextual evidence from large language models.**
Chaiho Shin, Dareen Eom, Hahyun You, Kwangsoo Kim, **Siun Kim***, Hyung-Jin Yoon*. (*co-corresponding) *BMC Medical Informatics and Decision Making*, 2026.
- Intra-session recurrence in intradialytic hypotension prediction: evaluation implications and recurrence-aware modeling.**
Siun Kim, Jiwon Ryu, Sejoong Kim, Su Hwan Kim, Myeongju Kim, Hyung-Jin Yoon. *Frontiers in Digital Health*, 2026.
- Beyond fine-tuning: leveraging domain-aware in-context learning with large language models for clinical named entity recognition.**
Siun Kim, David Seung U Lee, Yujin Kim, Hyung-Jin Yoon, Howard Lee. *Journal of Biomedical Informatics*, 2026.
- One-year post-acute COVID-19 syndrome and mortality in South Korea: a nationwide matched cohort study using claims data.**
Jung-Hyun Won, Yesol Hong, **Siun Kim**, Howard Lee. *Frontiers in Public Health*, 2024.
- Deep learning-based long-term risk evaluation of incident type 2 diabetes using electrocardiogram in a non-diabetic population: a retrospective, multicentre study.**
Junmo Kim, Hyun-Lim Yang, Su Hwan Kim, **Siun Kim**, Jisoo Lee, Jiwon Ryu, Kwangsoo Kim, Zio Kim, Gun Ahn, Doyun Kwon, Hyung-Jin Yoon. *eClinicalMedicine*, 2024.
- Automatic extraction of comprehensive drug safety information from adverse drug event narratives in the Korea Adverse Event Reporting System using natural language processing techniques.**
Siun Kim, Taegwan Kang, Tae Kyu Chung, Yoona Choi, YeSol Hong, Kyomin Jung, Howard Lee. *Drug Safety*, 2023.
- An annotated corpus from biomedical articles to construct a drug-food interaction database.**
Siun Kim, Yoona Choi, Jung-Hyun Won, Jung Mi Oh, Howard Lee. *Journal of Biomedical Informatics*, 2022.
- A population pharmacokinetic-pharmacodynamic model of YH12852, a highly selective 5-hydroxytryptamine 4 receptor agonist, in healthy subjects and patients with functional constipation.**
Siun Kim, Hyun A Lee, Seong Bok Jang, Howard Lee. *CPT: Pharmacometrics & Systems Pharmacology*, 2021.

9. **A critical review of the United States regulatory pathways for determining the equivalence of efficacy between CT-P13 and original infliximab (Remicade).**

Soohyun Kim, **Siun Kim**, Howard Lee. *Drug Design, Development and Therapy*, 2020.

Conference & Workshop Papers

10. **DiZiNER: disagreement-guided instruction refinement via simulating pilot annotation for zero-shot named entity recognition.**
Siun Kim, Hyung-Jin Yoon. *Proceedings of ACL 2026 (Main Conference)*.
11. **Questioning our questions: how well do medical QA benchmarks evaluate clinical capabilities of language models?**
Siun Kim, Hyung-Jin Yoon. *24th Workshop on Biomedical Language Processing (BioNLP @ ACL 2025) — Best Paper*.
12. **Large language model-assisted systematic review: validation based on Cochrane review data.**
Siun Kim, Hyung-Jin Yoon. *Medical Informatics Europe (MIE) 2025*.
13. **CRese: benchmark data and automatic evaluation framework for recommending eligibility criteria from clinical trial information.**
Siun Kim, Jung-Hyun Won, David Seung U Lee, Renqian Luo, Lijun Wu, Tao Qin, Howard Lee. *Findings of EACL 2024*.
14. **Revisiting machine learning based drug repurposing: drug indications are not a right prediction target.**
Siun Kim, Jung-Hyun Won, David Seung U Lee, Renqian Luo, Lijun Wu, Yingce Xia, Tao Qin, Howard Lee. *Conference on Health, Inference, and Learning (CHIL 2023)*, PMLR.

Book

15. **The era of biologics is coming.**
Siun Kim, Howard Lee. (Selected as 2022 Sejong Book) *Young Doc Press*, 2019.

PROJECTS

Selta Square

LLM agent system for pharmacovigilance case processing 2026.1 – Present

Designed an agent system that turns standard PV operating procedures into step-wise procedure modules, bounding the agent's autonomy with code-checked stage gates, tool permissions, enforced output formats and explicit human approval.

Causality assessment for adverse event cases 2026.1 – Present

Separated causality assessment, a largely self-contained step of case processing, into its own product and developed it first.

Document review for ICSR candidate extraction 2026.1 – Present

LLM-based document review that identifies potential individual case safety reports (ICSRs) in source documents.

Drug-label evidence retrieval for medical affairs 2026.1 – Present

Retrieval of established drug information (e.g., mechanism of action) from approved labels, with label selection prioritized by active ingredient, dosage form and strength, and handling of combination products.

Seoul National University Hospital

Zero-shot NER via disagreement signals across multiple LLMs 2023.3 – 2025.10

Label-free method that simulates pilot annotation, refining annotation instructions from disagreements among heterogeneous LLMs; reached zero-shot state of the art without instruction fine-tuning (ACL 2026 Main).

How well medical QA benchmarks reflect the clinical capabilities of LLMs 2025.2 – 2025.8

Systematically correlated LLM scores on major medical QA benchmarks (MedQA, MedMCQA, PubMedQA, MMLU medical subjects) with real-world clinical performance; benchmarks only partially reflect clinical capability.

In-context learning for clinical information extraction from hematology-oncology notes 2023.3 – 2025.10

Extracting diagnoses, symptoms, and prescriptions using few-shot retrieval and LLM in-context learning.

Intra-dialysis hypotension (IDH) prediction from TDMS time-series vital-sign and dialyzer data SNUH · 2023.3 –

2025.8 · Lead researcher

Recurrence-aware modeling and evaluation for long-term dialysis patients.

Seoul National University

Revisiting ML-based drug repositioning with clinical-trial eligibility criteria Microsoft Research Asia · 2022.5 – 2023.4

· Project lead

Showed that drug indications are not a sound prediction target for ML-based drug repositioning (CHIL 2023), and reframed the problem as recommending clinical-trial eligibility criteria with a public benchmark and an automatic evaluation framework (Findings of EACL 2024).

NLP for extracting drug safety information from adverse event reports Ministry of Food and Drug Safety · 2021.2 – 2022.2

· Project lead

KAERS-BERT for mixed Korean-English narratives; ICH E2B(R3)-based annotated corpus; NER, RE, sentence extraction and label classification models.

Drug-interaction evaluation using big data and machine learning Ministry of Food and Drug Safety · 2019.2 – 2022.2 ·

Subpart lead

NLP for drug-food interaction extraction and drug-drug interaction prediction.

PK/PD data analysis for Phase 1 trials of YH12852 and CWP232291 2018.3 – 2022.12

Analysis resulting in a population PK/PD model publication.

Prospective real-world registry for pediatric ALL from multi-institutional EHRs Korea Health Industry Development

Institute · 2022.3 – 2022.12

Report-form design and extraction-algorithm development.

IDX-1197 development strategy consultation Idience Co., Ltd · 2020.10 – 2021.2 · Lead researcher

Pre-clinical and clinical database construction for PARP inhibitors.

Target Product Profile (TPP) of REX101, a bacterial-exosome anticancer therapy Rosetta Exosome · 2021.3 – 2021.12 ·

Lead researcher

CONFERENCE PRESENTATIONS

CRESE: benchmark data and automatic evaluation framework for recommending eligibility criteria from clinical trial information — *EACL 2024, Malta*, Mar. 2024 · Poster

How to address data scarcity and bias for AI-powered drug discovery — *2023 Fall Korean Society of Medical Informatics Conference, Seoul*, Nov. 2023 · Oral

An annotated corpus and NLP model for drug safety information extraction from free-text adverse-event reports — *2021 Fall Korean Society of Medical Informatics Conference, Seoul*, Nov. 2021 · Oral

Building natural language training data for clinical information extraction — *2021 Korean Society of Medical Informatics Conference, Yongin*, Nov. 2021 · Tutorial

A population PK-PD model of YH12852, a highly selective 5-HT₄ receptor agonist — *Population Approach Group in Korea, Iksan*, Nov. 2020 · Oral

Development of an annotation guideline to extract drug-food interaction information for machine learning — *Korean Society for Clinical Pharmacology and Therapeutics, Yeosu*, Nov. 2019 · Poster

A population PK-PD model to characterize the increase in gastrointestinal motility by YH12852 — *2019 AAPS PharmSci 360, San Antonio, USA*, Nov. 2019 · Poster

AWARDS & HONORS

First Prize, Best Paper — The 24th Workshop on Biomedical Language Processing (BioNLP @ ACL 2025), Vienna, Austria, Aug. 2025

Best Article Presentation (1st) — Korean Society for Clinical Pharmacology and Therapeutics, Nov. 2021

Best Original Research Article (1st) — The Korean Society of Medical Informatics, Oct. 2021

Presidential Science Scholarship — Korea Student Aid Foundation, 2013 – 2017

PATENT

Apparatus and method for extracting drug-food interaction information — Howard Lee, **Siun Kim**, KR 10-2021-0172058 (application)

REVIEW EXPERIENCE

Conferences: NeurIPS (2026), ML4H (2024, 2025, 2026), MLHC (2025, 2026), BioNLP Workshop (2025)

Journal: JAMIA (2024, 2025, 2026)

SKILLS & LANGUAGES

Python, TypeScript, SQL, R, PyTorch, JAX, scikit-learn, pandas, NumPy, SciPy, Transformers, BERTopic, LangChain, Anthropic API, OpenAI API, OpenRouter, Ollama, Langfuse, FAISS, Claude Code, FastAPI, React, Streamlit, PostgreSQL, SQLite, Git, GitHub Actions, pytest, Docker, Amazon S3, Amazon ECS, Amazon EC2, AWS Lambda, Amazon CloudWatch, Amazon Bedrock, NONMEM

Languages: Korean (native), English (business fluent)